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Chapter 4 The First Fundamental Form

If V is a vector space, an inner product on V is a map that defines a pairing $\langle \mathbf{u}, \mathbf{w} \rangle$ between any two vectors $\mathbf{u}$ and $\mathbf{w}$ on V, which satisfies the following conditions.

(i) For any $\mathbf{u}$ and $\mathbf{w}$ in V,

$$\langle \mathbf{u}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{u} \rangle.$$

(ii) For any real numbers α_1, α_2, β_1, β_2, and any vectors $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{w}_1$, $\mathbf{w}_2$ in V,

$$\begin{aligned}&\langle \alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2, \beta_1 \mathbf{w}_1 + \beta_2 \mathbf{w}_2 \rangle \\ &= \alpha_1 \beta_1 \langle \mathbf{u}_1, \mathbf{w}_1 \rangle + \alpha_2 \beta_1 \langle \mathbf{u}_2, \mathbf{w}_1 \rangle + \alpha_1 \beta_2 \langle \mathbf{u}_1, \mathbf{w}_2 \rangle + \alpha_2 \beta_2 \langle \mathbf{u}_2, \mathbf{w}_2 \rangle.\end{aligned}$$

In other words, an inner product is a bilinear form.

(iii) For any $\mathbf{w}$ in V, $\langle \mathbf{w}, \mathbf{w} \rangle \geq 0$, and $\langle \mathbf{w}, \mathbf{w} \rangle = 0$ if and only if $\mathbf{w} = \mathbf{0}$.

The vector space $\mathbb{R}^3$ is equipped with the Euclidean inner product

$$\langle \mathbf{u}, \mathbf{w} \rangle = \mathbf{v} \cdot \mathbf{w} = u_1 w_1 + u_2 w_2 + u_3 w_3.$$

If W is a subspace of $\mathbb{R}^3$, then it inherits this inner product from $\mathbb{R}^3$, which is still a bilinear form on W.

If V_1 and V_2 are inner product spaces with inner products $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$, a map $T : V_1 \to V_2$ is said to be an inner product space isomorphism if $T : V_1 \to V_2$ is a bijective linear transformation, and for any $\mathbf{u}$ and $\mathbf{w}$ in V_1,

$$\langle T(\mathbf{u}), T(\mathbf{w}) \rangle_2 = \langle \mathbf{u}, \mathbf{w} \rangle_1.$$

4.1 Lengths of Curves on Surfaces

If S is a regular surface and $\mathbf{x}_0$ is a point on S, then the tangent space of S at $\mathbf{x}_0$, denoted by $T_{\mathbf{x}_0}S$, is a two dimensional subspace of $\mathbb{R}^3$. If $\boldsymbol{\sigma} : \mathcal{O} \to S \cap V$ is a coordinate patch such that $\boldsymbol{\sigma}(\mathbf{a}) = \mathbf{x}_0$, then the two vectors $\boldsymbol{\sigma}_u(\mathbf{a})$ and $\boldsymbol{\sigma}_v(\mathbf{a})$ is a basis of the vector space $T_{\mathbf{x}_0}S$. The restriction of the Euclidean inner product to $T_{\mathbf{x}_0}S$ is a bilinear form.

Definition 4.1 First Fundamental Form

Let S be a regular surface in $\mathbb{R}^3$ and let $\mathbf{x}_0$ be a point on S. The first fundamental form of S at $\mathbf{x}_0$ is the bilinear form $\langle \cdot, \cdot \rangle_{\mathbf{x}_0,S}$, which is the restriction of the Euclidean inner product to the tangent space $T_{\mathbf{x}_0}S$. For $\mathbf{v}$ and $\mathbf{w}$ in $T_{\mathbf{x}_0}S$,

$$\langle \mathbf{v}, \mathbf{w} \rangle_{\mathbf{x}_0,S} = \mathbf{v} \cdot \mathbf{w}.$$

Classically, the first fundamental form is defined as the quadratic form $Q_{\mathbf{x}_0} : T_{\mathbf{x}_0}S \to \mathbb{R}$,

$$Q_{\mathbf{x}_0}(\mathbf{w}) = \langle \mathbf{w}, \mathbf{w} \rangle_{\mathbf{x}_0,S}.$$

Given that $\boldsymbol{\sigma} : \mathcal{O} \to S$ is a coordinate patch with $\boldsymbol{\sigma}(\mathbf{a}) = \mathbf{x}_0$, the tangent space $T_{\mathbf{x}_0}S$ is spanned by the vectors $\boldsymbol{\sigma}_u(\mathbf{a})$ and $\boldsymbol{\sigma}_v(\mathbf{a})$. Any vector $\mathbf{w}$ in $T_{\mathbf{x}_0}S$ can be written uniquely as

$$\mathbf{w} = \alpha\boldsymbol{\sigma}_u(\mathbf{a}) + \beta\boldsymbol{\sigma}_v(\mathbf{a}).$$

Then

$$\begin{aligned}
&Q_{\mathbf{x}_0}(\mathbf{w}) \\
&= \alpha^2\langle\boldsymbol{\sigma}_u(\mathbf{a}), \boldsymbol{\sigma}_u(\mathbf{a})\rangle_{\mathbf{x}_0,S} + 2\alpha\beta\langle\boldsymbol{\sigma}_u(\mathbf{a}), \boldsymbol{\sigma}_v(\mathbf{a})\rangle_{\mathbf{x}_0,S} + \beta^2\langle\boldsymbol{\sigma}_v(\mathbf{a}), \boldsymbol{\sigma}_v(\mathbf{a})\rangle_{\mathbf{x}_0,S} \\
&= E\alpha^2 + 2\alpha\beta F + G\beta^2,
\end{aligned}$$

where

$$\begin{aligned}
E &= \langle\boldsymbol{\sigma}_u(\mathbf{a}), \boldsymbol{\sigma}_u(\mathbf{a})\rangle, \\
F &= \langle\boldsymbol{\sigma}_u(\mathbf{a}), \boldsymbol{\sigma}_v(\mathbf{a})\rangle, \\
G &= \langle\boldsymbol{\sigma}_v(\mathbf{a}), \boldsymbol{\sigma}_v(\mathbf{a})\rangle.
\end{aligned}$$

Define the linear functional $du : T_{\mathbf{x}_0}S \to \mathbb{R}$ and $dv : T_{\mathbf{x}_0}S \to \mathbb{R}$ by

$$du\left(\alpha\boldsymbol{\sigma}_u(\mathbf{a}) + \beta\boldsymbol{\sigma}_v(\mathbf{a})\right) = \alpha,$$

$$dv\left(\alpha\boldsymbol{\sigma}_u(\mathbf{a}) + \beta\boldsymbol{\sigma}_v(\mathbf{a})\right) = \beta.$$

Then we can write the first fundamental form $Q_{\mathbf{x}_0}$ in local coordinates as

$$Q_{\mathbf{x}_0} = Edu^2 + 2Fdudv + Gdv^2.$$

It is called the first fundamental form of the coordinate patch $\boldsymbol{\sigma} : \mathcal{O} \to S$ of S.

Let $\boldsymbol{\sigma} : \mathcal{O} \to S \cap V$ be a coordinate patch, and let $\boldsymbol{\gamma} : I \to S \cap V$ be a curve that lies on $S \cap V$. There is a map $\boldsymbol{\alpha} : I \to \mathcal{O}$,

$$\boldsymbol{\alpha}(t) = (u(t), v(t))$$

such that

$$\boldsymbol{\gamma}(t) = (\boldsymbol{\sigma} \circ \boldsymbol{\alpha})(t) = \boldsymbol{\sigma}(u(t), v(t)).$$

By chain rule,

$$\boldsymbol{\gamma}'(t) = u'(t)\boldsymbol{\sigma}_u(\boldsymbol{\alpha}(t)) + v'(t)\boldsymbol{\sigma}_v(\boldsymbol{\alpha}(t)).$$

It follows that

$$\|\boldsymbol{\gamma}'(t)\|^2 = Eu'(t)^2 + 2Fu'(t)v'(t) + Gv'(t)^2.$$

If $[a, b] \subset I$, the arc-length of the curve from $\boldsymbol{\gamma}(a)$ to $\boldsymbol{\gamma}(b)$ is

$$\int_a^b \sqrt{Eu'(t)^2 + 2Fu'(t)v'(t) + Gv'(t)^2}\, dt.$$

Example 4.1

Let $\mathbf{x}_0$ be a point in $\mathbb{R}^3$, and let $\mathbf{w}_1$ and $\mathbf{w}_2$ be two linearly independent vectors in $\mathbb{R}^3$. Consider the plane $\boldsymbol{\sigma} : \mathbb{R}^2 \to \mathbb{R}^3$,

$$\boldsymbol{\sigma}(u, v) = \mathbf{x}_0 + u\mathbf{w}_1 + v\mathbf{w}_2.$$

Find the first fundamental form of the plane in the coordinates (u, v).

Example 4.2

Let I be an open interval, and let $f : I \to \mathbb{R}$ be a smooth function with $f(x) > 0$ for all $x \in I$. The graph of f is a curve C in the xy-plane. Let S be the surface of revolution obtained by rotating C about the x-axis. Then a coordinate patch of S is given by $\boldsymbol{\sigma} : I \times J \to \mathbb{R}^3$,

$$\boldsymbol{\sigma}(u, v) = (u, f(u)\cos v, f(u)\sin v)\,,$$

where J is an open interval with length less than or equal to 2π. Find the first fundamental form of S in this local coordinates.

Example 4.3

A coordinate patch of the sphere $x^2 + y^2 + z^2 = 1$ is given by $\boldsymbol{\sigma} : I \times J \to \mathbb{R}^3$,

$$\boldsymbol{\sigma}(u, v) = (\sin u \cos v, \sin u \sin v, \cos u).$$

Here I is an open interval in $(0, \pi)$, and J is an open interval of length less than or equal to 2π. Find the first fundamental form of the sphere in this local coordinates.

Example 4.4

Let I be an open interval, and let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a curve parametrized by arc-length, lying in a plane whose unit normal vector is the vector $\mathbf{w}$. Consider the generalized cylinder S with parametrization $\boldsymbol{\sigma} : I \times \mathbb{R} \to \mathbb{R}^3$,

$$\boldsymbol{\sigma}(u, v) = \boldsymbol{\gamma}(u) + v\mathbf{w}.$$

Find the first fundamental form of S in the local coordinates (u, v).

Example 4.5

Let I be an open interval, and let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a curve that lies on the sphere $x^2 + y^2 + z^2 = 1$ and parametrized by arc-length. Consider the generalized cone S with parametrization $\boldsymbol{\sigma} : I \times (\mathbb{R} \setminus \{\mathbf{0}\}) \to \mathbb{R}^3$,

$$\boldsymbol{\sigma}(u, v) = v\boldsymbol{\gamma}(u).$$

Find the first fundamental form of S in the local coordinates (u, v).

Theorem 4.1

Let S be a surface and let $\boldsymbol{\sigma} : \mathcal{O} \to S$ be a coordinate patch of S. If $Edu^2 + 2Fdudv + Gdv^2$ is the first fundamental form of $\boldsymbol{\sigma}$, then

$$\begin{bmatrix} E & F \\ F & G \end{bmatrix} = \mathbf{D}\boldsymbol{\sigma}^T \mathbf{D}\boldsymbol{\sigma}.$$

Theorem 4.2

Let S be a surface and let $\boldsymbol{\sigma}_1 : \mathcal{O}_1 \to S$ and $\boldsymbol{\sigma}_2 : \mathcal{O}_2 \to S$ be coordinate patches of S. Assume that $\mathbf{\Phi}$ is the transition map such that $\boldsymbol{\sigma}_1 \circ \mathbf{\Phi} = \boldsymbol{\sigma}_2$. Let $E_1 du_1^2 + 2F_1 du_1 dv_1 + G_1 dv_1^2$ and $E_2 du_2^2 + 2F_2 du_2 dv_2 + G_2 dv_2^2$ be respectively the first fundamental forms of these two coordinate patches. Then

$$\begin{bmatrix} E_2 & F_2 \\ F_2 & G_2 \end{bmatrix} = \mathbf{D\Phi}^T \begin{bmatrix} E_1 & F_1 \\ F_1 & G_1 \end{bmatrix} \mathbf{D\Phi}.$$

4.2 Isometries of Surfaces

Consider the surface S_1 with paramentrization $\boldsymbol{\sigma}_1 : \mathbb{R}^2 \to \mathbb{R}^3$,

$$\boldsymbol{\sigma}_1(u, v) = (u, v, 0),$$

and the surface S_2 with parametrization $\boldsymbol{\sigma}_2 : (0, 2\pi) \times \mathbb{R} \to \mathbb{R}^3$,

$$\boldsymbol{\sigma}_2(u, v) = (\cos u, \sin u, v).$$

S_1 is the plane $z = 0$, while S_2 is part of the cylinder $x^2 + y^2 = 1$.

The first fundamental forms of $\boldsymbol{\sigma}_1 : \mathbb{R}^2 \to \mathbb{R}^3$ and $\boldsymbol{\sigma}_2 : (0, 2\pi) \times \mathbb{R} \to \mathbb{R}^3$ in the coordinates (u, v) are both given by

$$du^2 + dv^2.$$

This can be explained in the following way. We can ”wrap” a planar piece of paper on a cylinder without crumpling the paper. A curve on the plane will then become a curve on the cylinder. Because there is no crumpling, the lengths of these two curves are the same. This gives a heuristic reason why the first fundamental forms of these two surfaces are the same.

Definition 4.2 Isometry of Surfaces

Let S_1 and S_2 be surfaces in $\mathbb{R}^3$, and let $f : S_1 \to S_2$ be a map from S_1 to S_2. We say that f is an isometry provided that the following two conditions are satisfied.

(i) $f : S_1 \to S_2$ is a diffeomorphism.

(ii) For any $\mathbf{x}_0$ in S_1, the derivative map $D_{\mathbf{x}_0} f : T_{\mathbf{x}_0} S_1 \to T_{f(\mathbf{x}_0)} S_2$ is an inner product space isomorphism. Namely, for any $\mathbf{u}$ and $\mathbf{w}$ in $T_{\mathbf{x}_0} S_1$,

$$\langle D_{\mathbf{x}_0} f(\mathbf{u}), D_{\mathbf{x}_0} f(\mathbf{w}) \rangle_{f(\mathbf{x}_0), S_2} = \langle \mathbf{u}, \mathbf{w} \rangle_{\mathbf{x}_0, S_1}.$$

If there is an isometry $f : S_1 \to S_2$, we say that S_1 is isometric to S_2.

Definition 4.3 Local Isometry of Surfaces

Let S_1 and S_2 be surfaces in $\mathbb{R}^3$, and let $f : S_1 \to S_2$ be a map from S_1 to S_2. We say that f is a local isometry provided that for any $\mathbf{x}_0$ in S_1, the derivative map $D_{\mathbf{x}_0} f : T_{\mathbf{x}_0} S_1 \to T_{f(\mathbf{x}_0)} S_2$ is an inner product space isomorphism. Namely, for any $\mathbf{u}$ and $\mathbf{w}$ in $T_{\mathbf{x}_0} S_1$,

$$\langle D_{\mathbf{x}_0} f(\mathbf{u}), D_{\mathbf{x}_0} f(\mathbf{w}) \rangle_{f(\mathbf{x}_0), S_2} = \langle \mathbf{u}, \mathbf{w} \rangle_{\mathbf{x}_0, S_1}.$$

A local isometry is a local diffeomorphism. It is an isometry if and only if it is a diffeomorphism.

Definition 4.4 Locally Isometric

Let S_1 and S_2 be surfaces. We say that S_1 is locally isometric to S_2 at the point $\mathbf{x}_0$ if there exists a neighbourhood U of $\mathbf{x}_0$ in S_1 such that there is a local isometry $f : U \to S_2$.

We say that S_1 is locally isometric to S_2 if S_1 is locally isometric to S_2 at each point $\mathbf{x}_0$ of S_1.

If S_1 is locally isometric to S_2 and S_2 is locally isometric to S_1, then S_1 and S_2 are locally isometric.

If there exists a local isometry $f : S_1 \to S_2$, S_1 is locally isometric to S_2.

Proposition 4.3

Let S_1 and S_2 be surfaces, and let $f : S_1 \to S_2$ be a bijective smooth map. Then $f : S_1 \to S_2$ is a local isometry if and only if for any $\mathbf{x}_0$ in S_1, for any $\mathbf{u}$ in $T_{\mathbf{x}_0}S_1$,

$$\langle D_{\mathbf{x}_0} f(\mathbf{u}), D_{\mathbf{x}_0} f(\mathbf{u}) \rangle_{f(\mathbf{x}_0), S_2} = \langle \mathbf{u}, \mathbf{u} \rangle_{\mathbf{x}_0, S_1}.$$

Example 4.6

Let

$$S_1 = \{(x, y, z) \mid y = 0\}, \qquad S_2 = \left\{(x^2 + y^2, z) \mid x^2 + y^2 = 1\right\}.$$

The map $f : S_1 \to S_2$,

$$f(x, 0, z) = (\cos x, \sin x, z)$$

is a local isometry.

Proposition 4.4

Let S_1 and S_2 be surfaces, and let $f : S_1 \to S_2$ be an isometry. Then $f^{-1} : S_2 \to S_1$ is also an isometry.

This proposition says that if S_1 is isometric to S_2, then S_2 is isometric to S_1. Hence, if there is an isometry $f : S_1 \to S_2$, we say that S_1 and S_2 are isometric.

Proposition 4.5

Let S_1, S_2 and S_3 be surfaces, and let $f : S_1 \to S_2$ and $g : S_2 \to S_3$ be isometries. Then $(g \circ f) : S_1 \to S_3$ is also an isometry.

Corollary 4.6

Isometries of surfaces is an equivalence relation.

Proposition 4.7

Let S_1 and S_2 be surfaces, and let $f : S_1 \to S_2$ be a local isometry. Given a curve $\boldsymbol{\gamma}_1 : [a, b] \to S_1$ on S_1, let $\boldsymbol{\gamma}_2 : [a, b] \to S_2$ be the curve $\boldsymbol{\gamma}_2 = f \circ \boldsymbol{\gamma}_1$ on S_2. Then $\boldsymbol{\gamma}_1 : [a, b] \to S_1$ and $\boldsymbol{\gamma}_2 = f \circ \boldsymbol{\gamma}_1$ have the same length.

Lemma 4.8

Let V and W be two-dimensional vector spaces, and let $\{\mathbf{v}_1, \mathbf{v}_2\}$ be a basis of V. If $T : V \to W$ is a linear transformation, then $T : V \to W$ is an inner product space isomorphism if and only if

$$\langle T(\mathbf{v}_1), T(\mathbf{v}_1)\rangle = \langle \mathbf{v}_1, \mathbf{v}_1\rangle,$$

$$\langle T(\mathbf{v}_1), T(\mathbf{v}_2)\rangle = \langle \mathbf{v}_1, \mathbf{v}_2\rangle,$$

$$\langle T(\mathbf{v}_2), T(\mathbf{v}_2)\rangle = \langle \mathbf{v}_2, \mathbf{v}_2\rangle.$$

Theorem 4.9

Let S_1 and S_2 be regular surfaces in $\mathbb{R}^3$.

(a) Let $\mathcal{O}$ be an open subset of $\mathbb{R}^2$, and let $\boldsymbol{\sigma}_1 : \mathcal{O} \to S_1 \cap V_1$ and $\boldsymbol{\sigma}_2 : \mathcal{O} \to S_2 \cap V_2$ be coordinate patches on S_1 and S_2 such that

$$E_1 du^2 + 2F_1 dudv + G_1 dv^2 = E_2 du^2 + 2F_2 dudv + G_2 dv^2.$$

Then $f = (\boldsymbol{\sigma}_2 \circ \boldsymbol{\sigma}_1^{-1}) : S_1 \cap V_1 \to S_2 \cap V_2$ is an isometry.

(b) Conversely, if $f : S_1 \to S_2$ is an isometry, and $\boldsymbol{\sigma}_1 : \mathcal{O} \to S_1$ is a coordinate patch of S_1, then $\boldsymbol{\sigma}_2 = (f \circ \boldsymbol{\sigma}_1) : \mathcal{O} \to S_2$ is a coordinate patch of S_2. If $E_1 du^2 + 2F_1 dudv + G_1 dv^2$ and $E_2 du^2 + 2F_2 dudv + G_2 dv^2$ are the first fundamental forms of $\boldsymbol{\sigma}_1$ and $\boldsymbol{\sigma}_2$ respectively, then

$$E_1 du^2 + 2F_1 dudv + G_1 dv^2 = E_2 du^2 + 2F_2 dudv + G_2 dv^2.$$

Theorem 4.10

Let S_1 and S_2 be regular surfaces in $\mathbb{R}^3$. Then S_1 is locally isometric to S_2 at the point $\mathbf{x}_0$ in S_1 if and only if there are coordinate patches $\boldsymbol{\sigma}_1 : \mathcal{O} \to S_1$ and $\boldsymbol{\sigma}_2 : \mathcal{O} \to S_2$ such that $\mathbf{x}_0$ is in the image of $\boldsymbol{\sigma}_1$, and the first fundamental forms of $\boldsymbol{\sigma}_1$ and $\boldsymbol{\sigma}_2$ satisfy

$$E_1 du^2 + 2F_1 du dv + G_1 dv^2 = E_2 du^2 + 2F_2 du dv + G_2 dv^2.$$

Example 4.7

The cylinder $x^2 + y^2 = 1$ and the plane $z = 0$ are locally isometric.

A cylinder and a plane are not isometric.

Example 4.8

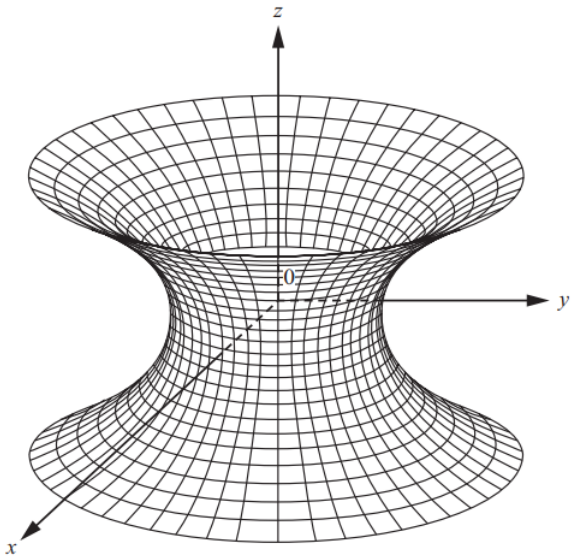
Let S_1 be the catenoid

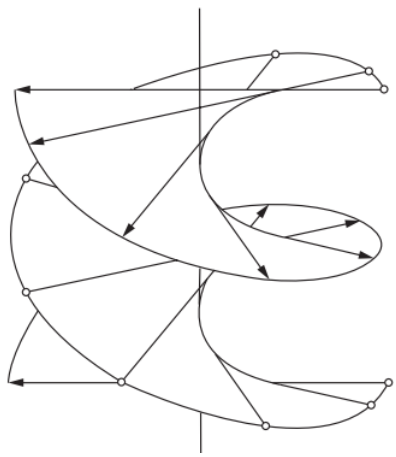
$$S_1 = \{(u, \cosh u \cos v, \cosh u \sin v) \mid u \in \mathbb{R}, v \in [0, 2\pi)\},$$

and let S_2 be the helicoid

$$S_2 = \{(v \cos u, v \sin u, u) \mid u \in \mathbb{R}, v \in \mathbb{R}\}.$$

Then S_1 and S_2 are locally isometric.





4.3 Conformal Mappings of Surfaces

Let S be a surface, and let $\boldsymbol{\gamma}_1 : I_1 \to S$ and $\boldsymbol{\gamma}_2 : I_2 \to S$ be two curves on S. If

$$\boldsymbol{\gamma}_1(t_1) = \boldsymbol{\gamma}_2(t_2) = \mathbf{x}_0,$$

then the angle θ between the two curves at the point $\mathbf{x}_0$ satisfies

$$\cos\theta = \frac{\langle \dot{\boldsymbol{\gamma}}_1(t_1), \dot{\boldsymbol{\gamma}}_2(t_2) \rangle}{\|\dot{\boldsymbol{\gamma}}_1(t_1)\| \, \|\dot{\boldsymbol{\gamma}}_2(t_2)\|}.$$

Definition 4.5 Conformal Maps

Let S_1 and S_2 be surfaces in $\mathbb{R}^3$, and let $f : S_1 \to S_2$ be a map from S_1 to S_2. We say that $f : S_1 \to S_2$ is a conformal map if it satisfies the following conditions:

(i) $f : S_1 \to S_2$ is a local diffeomorphism.

(ii) Given any $\mathbf{x}_0$ in S_1, if $\boldsymbol{\gamma}_1 : I_1 \to S_1$ and $\boldsymbol{\gamma}_2 : I_2 \to S_1$ are two curves on S_1 that pass through $\mathbf{x}_0$, and θ is the angle between these two curves at $\mathbf{x}_0$, then the angle between the two curves $(f \circ \boldsymbol{\gamma}_1) : I_1 \to S_2$ and $(f \circ \boldsymbol{\gamma}_2) : I_2 \to S_2$ at $f(\mathbf{x}_0)$ is also θ.

In other words, a conformal map is a map that preserves angles.

Theorem 4.11 Conformal Maps

Let S_1 and S_2 be surfaces in $\mathbb{R}^3$, and let $f : S_1 \to S_2$ be a local diffeomorphism. Then $f : S_1 \to S_2$ is a conformal map if and only if for any $\mathbf{x}_0 \in S_1$, for any $\mathbf{v}$ and $\mathbf{w}$ in $T_{\mathbf{x}_0}S_1$,

$$\frac{\langle D_{\mathbf{x}_0}f(\mathbf{v}), D_{\mathbf{x}_0}f(\mathbf{w})\rangle}{\|D_{\mathbf{x}_0}f(\mathbf{v})\|\,\|D_{\mathbf{x}_0}f(\mathbf{w})\|} = \frac{\langle \mathbf{v}, \mathbf{w}\rangle}{\|\mathbf{v}\|\,\|\mathbf{w}\|}.$$

Proposition 4.12

Let S_1 and S_2 be surfaces, and let $f : S_1 \to S_2$ be an isometry. Then $f : S_1 \to S_2$ is a conformal diffeomorphism.

Example 4.9

Let S be the plane $z = 0$, and let $f : S \to S$ be the map

$$f(\mathbf{x}) = c\mathbf{x},$$

where c is a nonzero constant. Then $f : S \to S$ is a conformal map. If $c \neq \pm 1$, $f : S \to S$ is not an isometry.

Proposition 4.13

Let S_1 and S_2 be surfaces, and let $f : S_1 \to S_2$ be a conformal diffeomorphism. Then $f^{-1} : S_2 \to S_1$ is also a conformal diffeomorphism.

Proposition 4.14

Let S_1, S_2 and S_3 be surfaces, and let $f : S_1 \to S_2$ and $g : S_2 \to S_3$ be conformal maps. Then $(g \circ f) : S_1 \to S_3$ is also a conformal map.

Theorem 4.15

Let S_1 and S_2 be surfaces, and let $f : S_1 \to S_2$ be a local diffeomorphism. Then $f : S_1 \to S_2$ is a conformal map if and only if there is a smooth function $\lambda : S_1 \to \mathbb{R}^+$ such that for any point $\mathbf{x}_0$ in S_1, and for any tangent vectors $\mathbf{v}$ and $\mathbf{w}$ in $T_{\mathbf{x}_0}S_1$,

$$\langle D_{\mathbf{x}_0} f(\mathbf{v}), D_{\mathbf{x}_0} f(\mathbf{w}) \rangle_{f(\mathbf{x}_0),S_2} = \lambda(\mathbf{x}_0) \langle \mathbf{v}, \mathbf{w} \rangle_{\mathbf{x}_0,S_1}.$$

Theorem 4.16

Let S_1 and S_2 be regular surfaces, and let $f : S_1 \to S_2$ be a local diffeomorphism. Then the followings are equivalent.

(a) The map $f : S_1 \to S_2$ is conformal.

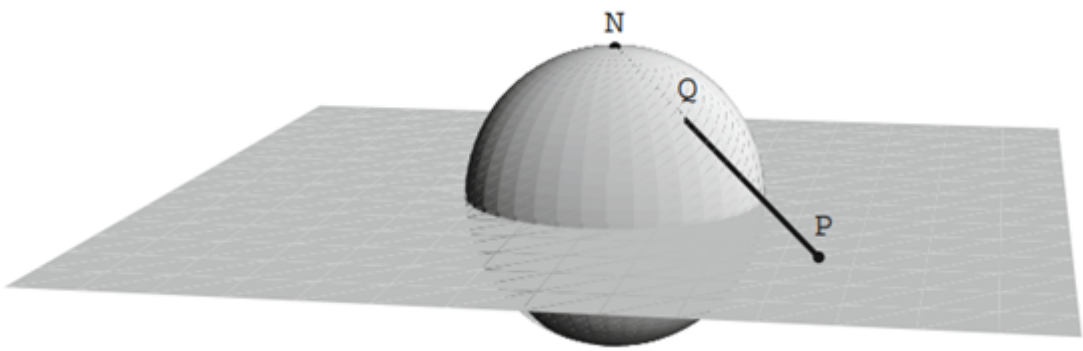
(b) Given that $\boldsymbol{\sigma} : \mathcal{O} \to S_1$ is a coordinate patch of S_1 such that $f|_{\boldsymbol{\sigma}(\mathcal{O})}$ is one-to-one, the first fundamental form $E_1 du^2 + 2F_1 dudv + G_1 dv^2$ of the coordinate patch $\boldsymbol{\sigma}$ of S_1 and the first fundamental form $E_2 du^2 + 2F_2 dudv + G_2 dv^2$ of the coordinate patch $f \circ \boldsymbol{\sigma}$ of S_2 are proportional. Namely, there is a smooth function $\lambda : \mathcal{O} \to \mathbb{R}$ such that

$$E_2 = \lambda E_1, \quad F_2 = \lambda F_1, \quad G_2 = \lambda G_1.$$

(c) There is an atlas of coordinate patches $\mathscr{A} = \{\boldsymbol{\sigma}_\alpha : \mathcal{O}_\alpha \to S_1 \,|\, \alpha \in J\}$ of S_1 such that for each $\alpha \in J$, $f|_{\boldsymbol{\sigma}_\alpha(\mathcal{O})}$ is one-to-one, and the first fundamental form $E_1 du^2 + 2F_1 dudv + G_1 dv^2$ of the coordinate patch $\boldsymbol{\sigma}_\alpha$ of S_1 and the first fundamental form $E_2 du^2 + 2F_2 dudv + G_2 dv^2$ of the coordinate patch $f \circ \boldsymbol{\sigma}_\alpha$ of S_2 are proportional.

Example 4.10 Stereographic Projection

Let S_1 be the unit sphere $x^2 + y^2 + z^2 = 1$ minus the north pole $\mathbf{p} = (0, 0, 1)$, and let S_2 be the plane $z = 0$. Given $\mathbf{x}_0$ on S_1, let L be the line segment joining $\mathbf{x}_0$ to $\mathbf{p}$. The stereographic projection is the map $f : S_1 \to S_2$ that takes the point $\mathbf{x}_0$ on S_1 to the point $f(\mathbf{x}_0)$, which is the intersection of the line L with the plane $z = 0$. Show that $f : S_1 \to S_2$ is conformal.



Definition 4.6 Conformal Patches

Let S be a regular surface, and let $\boldsymbol{\sigma} : \mathcal{O} \to S \cap V$ be a coordinate patch of S. We say that $\boldsymbol{\sigma} : \mathcal{O} \to S \cap V$ is a conformal patch of S if $\boldsymbol{\sigma} : \mathcal{O} \to S \cap V$ is a conformal diffeomorphism. Equivalent, if $Edu^2 + 2Fdudv + Gdv^2$ is the first fundamental form of $\boldsymbol{\sigma}$, then

$$F = 0, \qquad E = G.$$

Theorem 4.17

Every surface has an atlas consisting of conformal surface patches.

Definition 4.7 Locally Conformal

Let S_1 and S_2 be surfaces in $\mathbb{R}^3$, and let U be an open subset of S_1 that contains the point $\mathbf{x}_0$. We say that $f : U \to S_2$ is locally conformal at $\mathbf{x}_0 \in S_1$ provided that $f(U)$ is an open subset of S_2 and $f : U \to f(U)$ is a conformal map.

If there is a locally conformal map from S_1 to S_2 at every $\mathbf{x}_0 \in S_1$, we say that S_1 is locally conformal to S_2.

Proposition 4.18

Let S_1 and S_2 be surfaces. If S_1 is locally isometric to S_2, then it is locally conformal to S_2.

Theorem 4.19

Any two surfaces are locally conformal.

4.4 Equiareal Maps and a Theorem of Archemedes

Proposition 4.20 Change of Variables Theorem

Let $\mathcal{O}$ and $\mathcal{U}$ be open subsets of $\mathbb{R}^2$, and let $\mathbf{\Psi} : \mathcal{O} \to \mathcal{U}$ be a diffeomorphism. If $\mathfrak{D}$ is a Jordan measurable set such that its closure $\overline{\mathfrak{D}}$ is contained in $\mathcal{O}$, then $\mathbf{\Psi}(\mathfrak{D})$ is also Jordan measurable. If $f : \mathbf{\Psi}(\mathfrak{D}) \to \mathbb{R}$ is a bounded continuous function, then

$$\int_{\mathbf{\Psi}(\mathfrak{D})} f(\mathbf{x}) d\mathbf{x} = \int_{\mathfrak{D}} f\left(\mathbf{\Psi}(\mathbf{x})\right) \left|\det \mathbf{D}\mathbf{\Psi}(\mathbf{x})\right| d\mathbf{x}.$$

A bounded set $\mathfrak{D} \subset \mathbb{R}^n$ is Jordan measurable if the function $\chi_{\mathfrak{D}} : \mathbb{R}^n \to \mathbb{R}$,

$$\chi_{\mathfrak{D}}(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in \mathfrak{D}, \\ 0, & \text{otherwise}, \end{cases}$$

is Riemann integrable.

Lemma 4.21

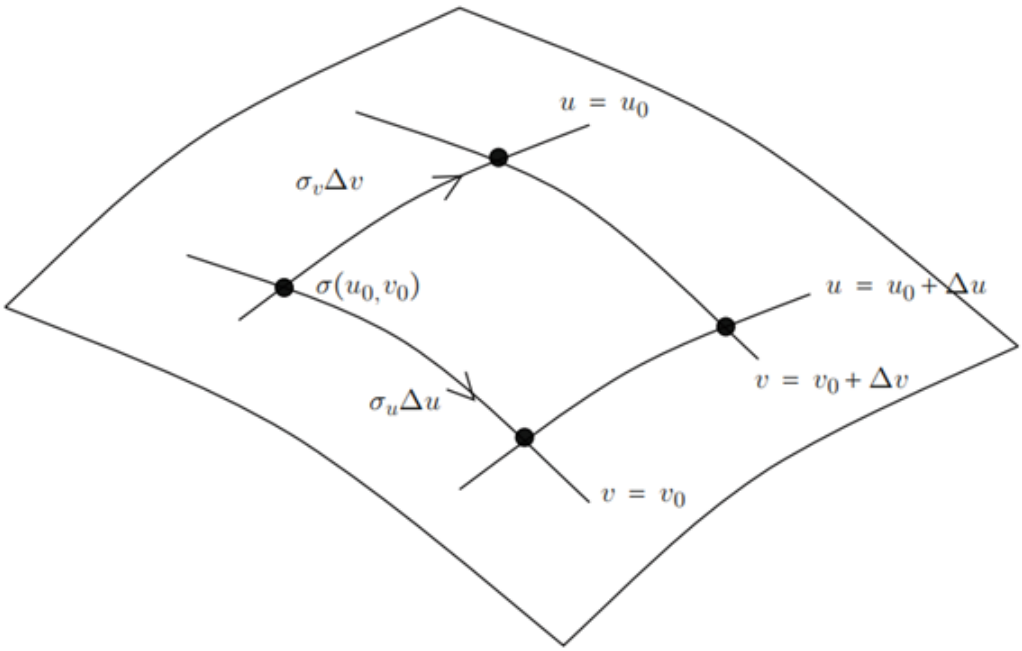
Let $\mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbb{R}^3$. The area of the parallelogram spanned by $\mathbf{v}$ and $\mathbf{w}$ is

$$\|\mathbf{v} \times \mathbf{w}\|.$$

Definition 4.8 Area of a surface

Let S be a surface, and let $\boldsymbol{\sigma} : \mathcal{O} \to S$ be a coordinate patch of S. If R is a Jordan measurable subset of $\mathcal{O}$, then the area of the part $\boldsymbol{\sigma}(R)$ of the surface S is

$$\mathcal{A}(R) = \int_R \|\boldsymbol{\sigma}_u \times \boldsymbol{\sigma}_v\| du dv.$$



Theorem 4.22

Let S be a surface and let $\boldsymbol{\sigma} : \mathcal{O} \to S$ be a coordinate patch of S with first fundamental form $Edu^2 + 2Fdudv + Gdv^2$. Then $EG - F^2 > 0$ and

$$\|\boldsymbol{\sigma}_u \times \boldsymbol{\sigma}_v\| = \sqrt{EG - F^2}.$$

If $\mathbf{\Phi} : \widetilde{\mathcal{O}} \to \mathcal{O}$ is a diffeomorphism, then $\widetilde{\boldsymbol{\sigma}} = \boldsymbol{\sigma} \circ \mathbf{\Phi} : \widetilde{\mathcal{O}} \to S$ is another coordinate patch of S. If $\widetilde{E}d\widetilde{u}^2 + 2\widetilde{F}d\widetilde{u}d\widetilde{v} + \widetilde{G}d\widetilde{v}^2$ is the first fundamental form of $\widetilde{\boldsymbol{\sigma}} : \widetilde{\mathcal{O}} \to S$, then

$$\widetilde{E}\widetilde{G} - \widetilde{F}^2 = (EG - F^2)\left(\det \mathbf{D}\mathbf{\Phi}\right)^2.$$

Theorem 4.23

Let S be a surface, and let $\boldsymbol{\sigma} : \mathcal{O} \to S$ be a coordinate patch of S. If R is a Jordan measurable subset of $\mathcal{O}$, then the area of the part $\boldsymbol{\sigma}(R)$ of the surface S is

$$\mathcal{A}(R) = \int_R \sqrt{EG - F^2} du dv.$$

It is independent of parametrizations.

Example 4.11

Let S the part of the unit sphere $x^2 + y^2 + z^2 = 1$ bounded between the planes $y = 0$ and $y = x \tan \alpha$, where α is a fixed number satisfying $0 < \alpha < \pi$. Find the area of S.

Definition 4.9 Equiareal Maps

Let S_1 and S_2 be surfaces in $\mathbb{R}^3$, and let $f : S_1 \to S_2$ be a map from S_1 to S_2. We say that f is an equiareal map provided that the following two conditions are satisfied.

(i) $f : S_1 \to S_2$ is a local diffeomorphism.

(ii) For any measurable region D_1 in S_1, $D_2 = f(D_1)$ is a measurable region in S_2 that has the same area as D_1.

Theorem 4.24

Let S_1 and S_2 be regular surfaces, and let $f : S_1 \to S_2$ be a local diffeomorphism. Then the followings are equivalent.

(a) The map $f : S_1 \to S_2$ is equiareal.

(b) Given that $\boldsymbol{\sigma} : \mathcal{O} \to S_1$ is a coordinate patch of S_1 such that $f|_{\boldsymbol{\sigma}(\mathcal{O})}$ is one-to-one, if the first fundamental form of the coordinate patch $\boldsymbol{\sigma}$ of S_1 is $E_1 du^2 + 2F_1 dudv + G_1 dv^2$, and the first fundamental form of the coordinate patch $f \circ \boldsymbol{\sigma}$ of S_2 is $E_2 du^2 + 2F_2 dudv + G_2 dv^2$, then

$$E_1 G_1 - F_1^2 = E_2 G_2 - F_2^2.$$

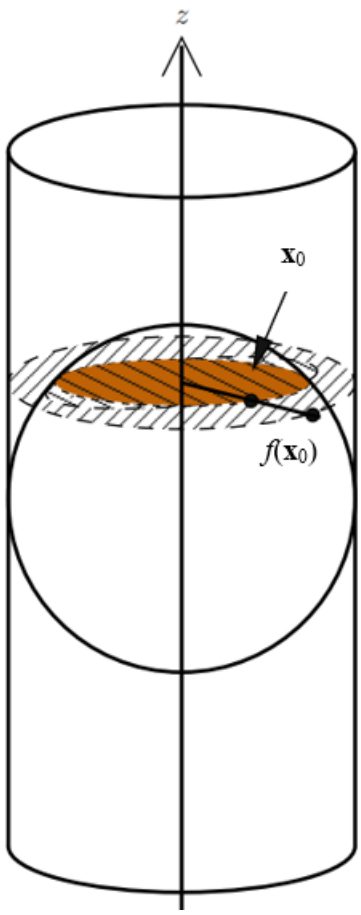
(c) There is an atlas of coordinate patches $\mathscr{A} = \{\boldsymbol{\sigma}_\alpha : \mathcal{O}_\alpha \to S_1 \,|\, \alpha \in J\}$ of S_1 such that for each $\alpha \in J$, $f|_{\boldsymbol{\sigma}_\alpha(\mathcal{O})}$ is one-to-one, and if the first fundamental form of the coordinate patch $\boldsymbol{\sigma}_\alpha$ of S_1 is $E_1 du^2 + 2F_1 dudv + G_1 dv^2$, the first fundamental form of the coordinate patch $f \circ \boldsymbol{\sigma}_\alpha$ of S_2 is $E_2 du^2 + 2F_2 dudv + G_2 dv^2$, then

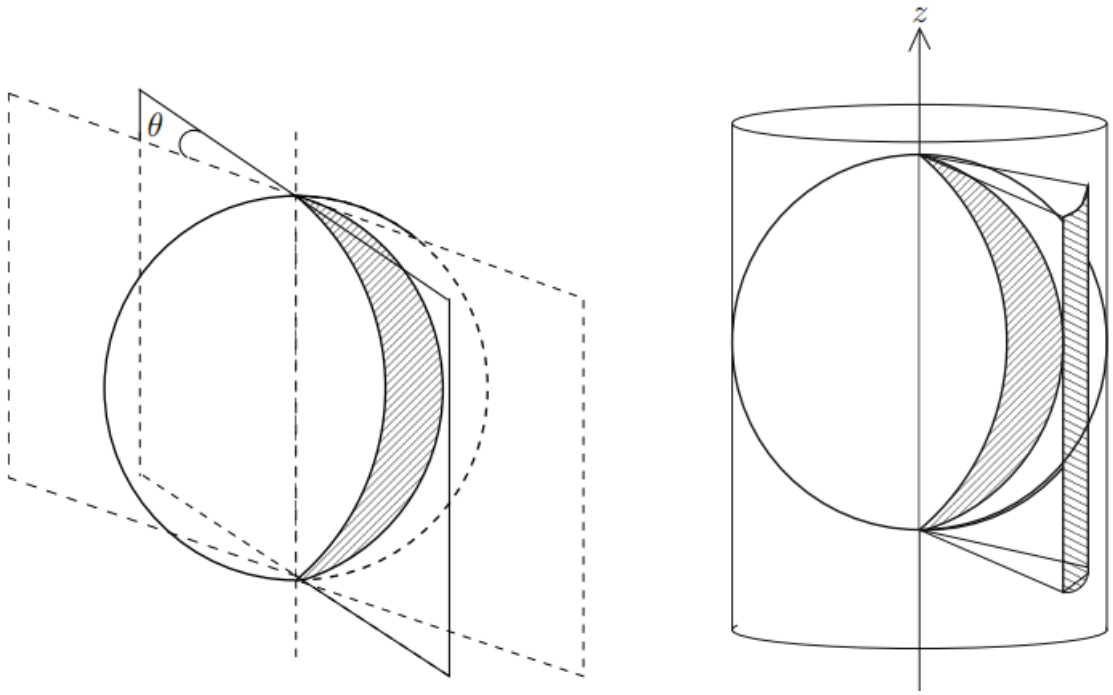
$$E_1 G_1 - F_1^2 = E_2 G_2 - F_2^2.$$

Archimedes has found one of the most famous examples of equiareal maps. Accordingly, this discovery was inscribed onto his tombstone.

Theorem 4.25 Archimedes' Equiareal Theorem

Let S_1 be the unit sphere $x^2 + y^2 + z^2 = 1$, and let S_2 be the cylinder $x^2 + y^2 = 1$. Let $\mathbf{p} = (0, 0, 1)$ and $\mathbf{q} = (0, 0, -1)$ be the north and south poles of the sphere. We define a map $f : S_1 \setminus \{\mathbf{p}, \mathbf{q}\} \to S_2$ in the following way. For any $\mathbf{x}_0 = (x_0, y_0, z_0) \in S_1 \setminus \{\mathbf{p}, \mathbf{q}\}$, let L be the straightline that passes through the point $\mathbf{x}_0$ and $(0, 0, z_0)$. Then L intersections the cylinder S_2 in two different points. Define $f(\mathbf{x}_0)$ to be the point that is closer to $\mathbf{x}_0$. Then f is an equiareal map.




θ
z

4.5 Spherical Geometry

Spherical geometry is important because we live on the surface of the earth, which is approximately a sphere.

It is sufficient for us to consider the unit sphere $x^2 + y^2 + z^2 = 1$. A great circle on the sphere is a circle on the sphere which lies on a plane that passes through the center of the sphere. A spherical triangle is a triangle on the sphere whose sides are arcs of great circles.

Proposition 4.26

Let S be the unit sphere $x^2 + y^2 + z^2 = 1$. Given that $\mathbf{x}_1$ and $\mathbf{x}_2$ are two distinct points on S, there is a great circle that passes through these two points.

Theorem 4.27

Let S be the unit sphere $x^2 + y^2 + z^2 = 1$. Given that $\mathbf{x}_1$ and $\mathbf{x}_2$ are two distinct points on S, among the curves on S that joins $\mathbf{x}_1$ to $\mathbf{x}_2$, the one that has the shortest length must lie on a great circle.

Hence, the great circles on spheres are analogues of straightlines on the plane. The main difference between spherical geometry and Euclidean geometry on the plane is that no lines are parallel in the spherical geometry.

Corollary 4.28

Given two distinct points $\mathbf{x}_1$ and $\mathbf{x}_2$ on the unit sphere S^2, the distance between these two points on the sphere is

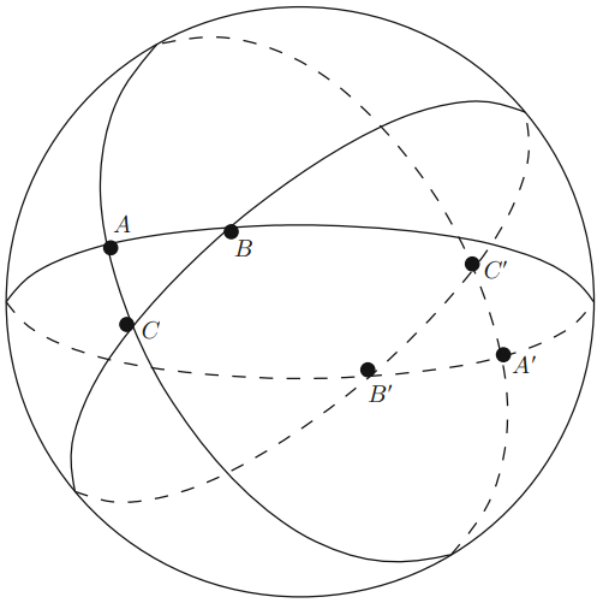
$$d_{S^2}(\mathbf{x}_1, \mathbf{x}_2) = \cos^{-1} \langle \mathbf{x}_1, \mathbf{x}_2 \rangle .$$

A spherical triangle on S^2 is a triangle whose edges are parts of great circles.

Theorem 4.29

The area of a spherical triangle on the unit sphere S^2 with interior angles α, β and γ is

$$\alpha + \beta + \gamma - \pi.$$



Definition 4.10 Reflections

A linear transformation $\mathbf{T_A} : \mathbb{R}^n \to \mathbb{R}^n$ is a reflection if there is a $(n-1)$-dimensional subspace W of $\mathbb{R}^n$ such that $\mathbf{T_A}(\mathbf{x}) = \mathbf{x}$ for all $\mathbf{x} \in W$, and $\mathbf{T_A}(\mathbf{n}) = -\mathbf{n}$ if $\mathbf{n}$ is a vector orthogonal to W.

Theorem 4.30

A linear transformation $\mathbf{T_A} : \mathbb{R}^n \to \mathbb{R}^n$ is a reflection if and only if $\mathbf{A}$ is a matrix that is orthogonally similar to a diagonal matrix with $(n-1)$ diagonal entries equal to 1, and one diagonal entry equal to -1.

Corollary 4.31

Let $\mathbf{T}_{\mathbf{A}} : \mathbb{R}^n \to \mathbb{R}^n$ be a reflection defined by the matrix $\mathbf{A}$. Then we have the following.

(a) The matrix $\mathbf{A}$ is symmetric and orthogonal.

(b) $\mathbf{A}^2 = \mathbf{I}_n$.

(c) $\det \mathbf{A} = -1$.

Lemma 4.32

Let θ be a real number. If L is the line $x \sin\theta = y \cos\theta$ in $\mathbb{R}^2$, $\mathbf{A}$ is the orthogonal matrix that defines the reflection in $\mathbb{R}^2$ with respect to L, then

$$\mathbf{A} = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}.$$

Theorem 4.33 Cartan-Dieudonné Theorem

Let n be an integer larger than or equal to 2. Then every orthogonal transformation of $\mathbb{R}^n$ is a composition of at most n reflections.